

MLA0201 – Fundamentals of Machine Learning
Analytical Problem Solving

Q1. Linear Regression (5 Marks – 10 Mins)

A real estate company collected the following training data:

House No	Area (X_1) (sq.ft)	Bedrooms (X_2)	Price (Y) (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct the Linear Regression Model

The multiple linear regression equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

where Y = House Price (₹ Lakhs), X_1 = Area (sq.ft), X_2 = Number of Bedrooms, β_0 = Intercept, β_1 , β_2 = Regression coefficients.

(ii) Estimate the Regression Coefficients

Step 1: Calculate the Means

Mean of Area: $\bar{X}_1 = (1000+1500+800+1200+2000) / 5 = 6500 / 5 = 1300$

Mean of Bedrooms: $\bar{X}_2 = (2+3+2+3+4) / 5 = 14 / 5 = 2.8$

Mean of Price: $\bar{Y} = (50+75+40+60+90) / 5 = 315 / 5 = 63$

Step 2: Estimate Regression Coefficients (using least squares approximation):

Area coefficient: $\beta_1 \approx 0.041 \rightarrow$ every additional 1 sq.ft increases the price by about ₹0.041 lakh.

Bedroom coefficient: $\beta_2 \approx 4.5 \rightarrow$ each additional bedroom increases the price by about ₹4.5 lakhs.

Intercept: $\beta_0 = \bar{Y} - \beta_1 \bar{X}_1 - \beta_2 \bar{X}_2 = 63 - (0.041 \times 1300) - (4.5 \times 2.8) = 63 - 53.3 - 12.6 \approx -2.9$

Estimated Regression Equation:

$$Y = -2.9 + 0.041X_1 + 4.5X_2$$

(iii) Interpretation of Coefficients

$\beta_0 = -2.9$: This is the intercept — the estimated house price when area and bedrooms are both zero. It has little practical meaning here.

$\beta_1 = 0.041$: Every additional 1 sq.ft increases the house price by approximately ₹0.041 lakh. Example: 100 extra sq.ft $\rightarrow 100 \times 0.041 = ₹4.1$ lakhs increase.

$\beta_2 = 4.5$: Every additional bedroom increases the house price by approximately ₹4.5 lakhs, assuming area remains the same.

Overall: Area and bedrooms both positively affect house price — larger houses with more bedrooms generally have higher prices.

Q2. Linear Classification (5 Marks – 10 Mins)

An email system uses the following feature values:

Email	Offer (X_1)	Win (X_2)	Spam (Y)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

(i) Construct the Linear Classification Model

A linear classifier has the form: $f(X) = w_0 + w_1X_1 + w_2X_2$

Choosing approximate weights based on the dataset: $w_0 = -2$, $w_1 = 1$, $w_2 = 1$

$$f(X) = -2 + X_1 + X_2$$

(ii) Determine the Decision Boundary

Decision boundary occurs when $f(X) = 0$:

$$-2 + X_1 + X_2 = 0 \Rightarrow X_1 + X_2 = 2$$

Classification rule: If $X_1 + X_2 > 2 \rightarrow \text{Spam} = 1$; otherwise $\rightarrow \text{Spam} = 0$

(iii) Classify New Email (Offer = 2, Win = 1)

$$f(X) = -2 + 2 + 1 = 1$$

Since $1 > 0$, the email belongs to the Spam class.

Q3. Naïve Bayes Classification

A medical dataset is given:

Patient	Fever (1/0)	Headache (1/0)	Disease
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute Prior Probabilities

Total patients = 5; Disease = Yes $\rightarrow$ 3; Disease = No $\rightarrow$ 2

$$P(\text{Yes}) = 3/5 = 0.6 \quad P(\text{No}) = 2/5 = 0.4$$

(ii) Compute Conditional Probabilities

For Disease = Yes (3 cases):

$$P(\text{Fever} = 1 \mid \text{Yes}) = 3/3 = 1$$

$$P(\text{Headache} = 0 \mid \text{Yes}) = 1/3$$

For Disease = No (2 cases):

$$P(\text{Fever} = 1 \mid \text{No}) = 0/2 = 0$$

$$P(\text{Headache} = 0 \mid \text{No}) = 1/2$$

(iii) Predict for Fever = 1, Headache = 0

Probability of Yes:

$$P(\text{Yes} \mid X) \propto P(\text{Yes}) \times P(\text{Fever}=1|\text{Yes}) \times P(\text{Headache}=0|\text{Yes}) = 0.6 \times 1 \times 1/3 = 0.2$$

Probability of No:

$$P(\text{No} \mid X) \propto P(\text{No}) \times P(\text{Fever}=1|\text{No}) \times P(\text{Headache}=0|\text{No}) = 0.4 \times 0 \times 1/2 = 0$$

Since $0.2 > 0$, the patient is classified as:

Q4. Linear Discriminant Analysis

Given dataset:

Data Point	X_1	X_2	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct the Discriminant Function

$$g(X) = w_0 + w_1X_1 + w_2X_2$$

$$\text{Mean of Class C1: } \mu_1 = ((2+3)/2, (3+4)/2) = (2.5, 3.5)$$

$$\text{Mean of Class C2: } \mu_2 = ((5+6+4)/3, (6+7+5)/3) = (5, 6)$$

The approximate discriminant function is:

$$g(X) = -8 + X_1 + X_2$$

(ii) Determine the Decision Boundary

$$\text{Decision boundary occurs when } g(X) = 0: -8 + X_1 + X_2 = 0 \Rightarrow X_1 + X_2 = 8$$

Classification rule: If $X_1 + X_2 < 8 \rightarrow \text{Class} = \text{C1}$; If $X_1 + X_2 \geq 8 \rightarrow \text{Class} = \text{C2}$

(iii) Classify New Point ($X_1 = 3, X_2 = 3$)

$$g(X) = -8 + 3 + 3 = -2$$

Since $g(X) < 0$, the point belongs to:

Class C1

Q5. Bayesian Logistic Regression

A telecom company collected the following data:

Customer	Monthly Charges (X ₁)	Tenure (X ₂)	Churn (Y)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

(i) Formulate the Bayesian Logistic Regression Model

The logistic regression equation is: $P(Y=1) = 1 / (1 + e^{-Z})$, where $Z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

Bayesian logistic regression estimates the coefficients using prior distributions.

(ii) Estimate Model Parameters

Approximate estimated coefficients: $\beta_0 = -5$, $\beta_1 = 0.01$, $\beta_2 = -0.20$

$$P(Y=1) = 1 / (1 + e^{-(-5 + 0.01X_1 - 0.20X_2)})$$

(iii) Predict Churn (Monthly Charges = 650, Tenure = 9)

Step 1: $Z = -5 + (0.01)(650) - (0.20)(9) = -5 + 6.5 - 1.8 = -0.3$

Step 2: $P = 1 / (1 + e^{0.3})$; $e^{0.3} \approx 1.35$; $P = 1 / 2.35 \approx 0.43$

Since $0.43 < 0.5$, the customer is predicted as:

No Churn (Customer will NOT churn)

Q6. k-Nearest Neighbour, k-NN

Given the following dataset for customer classification:

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

For a new customer (Age = 32, Income = 50):

(i) Compute Euclidean Distance from All Points

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Distance from C1 = $\sqrt{(32-25)^2 + (50-40)^2} = \sqrt{49+100} = \sqrt{149} = 12.21$

Distance from C2 = $\sqrt{(32-30)^2 + (50-60)^2} = \sqrt{4+100} = \sqrt{104} = 10.20$

Distance from C3 = $\sqrt{(32-35)^2 + (50-80)^2} = \sqrt{9+900} = \sqrt{909} = 30.15$

Distance from C4 = $\sqrt{(32-40)^2 + (50-20)^2} = \sqrt{64+900} = \sqrt{964} = 31.05$

(ii) Classify Using k = 3

Customer	Distance	Class
C2	10.20	A
C1	12.21	A
C3	30.15	B

Nearest 3 neighbours: C2 → A, C1 → A, C3 → B

Votes: A = 2, B = 1

(iii) Final Class Label

Majority class wins.

Q7. Decision Tree – Entropy & Information Gain

Given dataset:

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) Compute Entropy of Play

Classes: Yes = 2, No = 2, Total = 4 $\rightarrow P(\text{Yes}) = 0.5, P(\text{No}) = 0.5$

Entropy(S) = $-P(\text{Yes})\log_2 P(\text{Yes}) - P(\text{No})\log_2 P(\text{No}) = -0.5(-1) - 0.5(-1) = 0.5 + 0.5 = 1$

(ii) Calculate Information Gain for Outlook

Sunny $\rightarrow$ Play: No, No $\rightarrow$ Entropy = 0 (all one class)

Overcast $\rightarrow$ Play: Yes $\rightarrow$ Entropy = 0

Rain $\rightarrow$ Play: Yes $\rightarrow$ Entropy = 0

Weighted Entropy = $(2/4)(0) + (1/4)(0) + (1/4)(0) = 0$

Information Gain(Outlook) = Entropy(S) – Weighted Entropy = $1 - 0 = 1$

(iii) Identify the Root Node

The attribute with the highest Information Gain becomes the root node.

Q8. Support Vector Machine – Linear Hyperplane (5 Marks – 10 Mins)

Given data points:

X_1	X_2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

(i) Plot the Points

Positive points (+1): (2,2), (4,4) Negative points (-1): (2,4), (4,2)

(When plotted on the X_1 - X_2 plane, the positive points lie on the line $X_1 = X_2$, and the negative points lie symmetrically on either side of it.)

(ii) Determine a Linear Separating Hyperplane

A simple separating function is:

$$g(X) = X_1 - X_2$$

(iii) Find the Equation of the Decision Boundary

Set $g(X) = 0$: $X_1 - X_2 = 0 \Rightarrow X_1 = X_2$

Classification rule: If $X_1 > X_2 \rightarrow \text{Class} = +1$; If $X_1 < X_2 \rightarrow \text{Class} = -1$

$$\text{Decision Boundary: } X_1 = X_2$$

Q9. Confusion Matrix Evaluation

Logistic Regression

	Pred +	Pred -
Actual +	40	10
Actual -	5	45

TP = 40, FP = 5, FN = 10, TN = 45

Accuracy = $(TP+TN)/\text{Total} = (40+45)/100 = 0.85 = 85\%$

Precision = $TP/(TP+FP) = 40/45 = 0.889 = 88.9\%$

Recall = $TP/(TP+FN) = 40/50 = 0.80 = 80\%$

Naïve Bayes

	Pred +	Pred -
Actual +	35	15
Actual -	10	40

TP = 35, FP = 10, FN = 15, TN = 40

Accuracy = $(35+40)/100 = 0.75 = 75\%$

Precision = $35/45 = 0.778 = 77.8\%$

Recall = $35/50 = 0.70 = 70\%$

(i) & (ii) Comparison of Results

Metric	Logistic Regression	Naïve Bayes
Accuracy	85%	75%
Precision	88.9%	77.8%
Recall	80%	70%

Since Logistic Regression has higher Accuracy, Precision, and Recall:

Logistic Regression is the better model
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Q10. K-Means Clustering

Given points: (2,4), (3,5), (8,9), (9,10)

Initial centroids: $C_1 = (2,4)$, $C_2 = (8,9)$

(i) Perform One Iteration

Point (2,4): Distance to $C_1 = 0$; Distance to $C_2 = \sqrt{[(8-2)^2 + (9-4)^2]} = \sqrt{61} = 7.81 \rightarrow$ Assigned to C_1

Point (3,5): Distance to $C_1 = \sqrt{[1^2 + 1^2]} = 1.41$; Distance to $C_2 = \sqrt{[25 + 16]} = \sqrt{41} \dots = 6.40 \rightarrow$ Assigned to C_1

Point (8,9): Distance to $C_1 = 7.81$; Distance to $C_2 = 0 \rightarrow$ Assigned to C_2

Point (9,10): Distance to $C_1 = \sqrt{[7^2 + 6^2]} = \sqrt{85} = 9.22$; Distance to $C_2 = \sqrt{[1^2 + 1^2]} = 1.41 \rightarrow$ Assigned to C_2

(ii) Cluster Assignment

Cluster 1	Cluster 2
(2, 4)	(8, 9)
(3, 5)	(9, 10)

(iii) Compute New Centroids

New Centroid $C_1 = ((2+3)/2, (4+5)/2) = (2.5, 4.5)$

New Centroid $C_2 = ((8+9)/2, (9+10)/2) = (8.5, 9.5)$

$C_1 = (2.5, 4.5)$ $C_2 = (8.5, 9.5)$
